

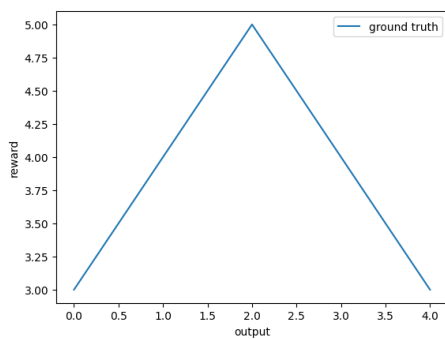
CS 521: Homework 4

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Due: November 7, 2025

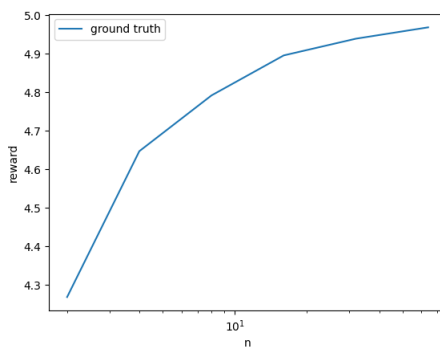
Link to files: <https://github.com/jfulfo/cs521/tree/main/hw4>

Problem 1

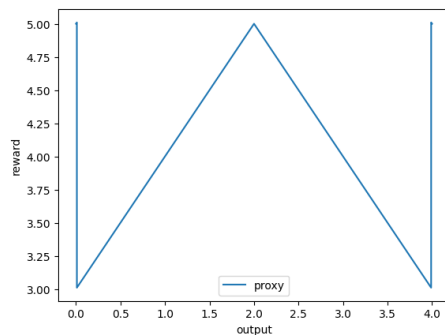


(a)

- (b) Expected reward increases monotonically with n . As n increases, best-of- N sampling is more likely to find samples closer to the optimal value $x = 2$, resulting in higher expected rewards. The improvement rate decreases as N grows larger due to diminishing returns.



Problem 2

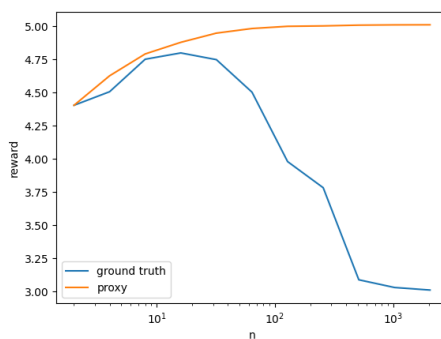


(a)

(b) Using formula $KL(n) = \log(n) - \frac{n-1}{n}$:

N	KL Divergence
1	0.0000
2	0.1931
4	0.6363
8	1.2044
16	1.8351
32	2.4970
64	3.1745

(c) There is clear reward hacking. The proxy reward increases monotonically and plateaus near 5.0, but the ground truth reward initially increases but peaks around $N=16$, then decreases. This occurs because a large N causes the model to exploit the proxy's flaws (artificial +2 bonus at boundaries). A large N is bad because it strengthens reward hacking by increasingly exploiting proxy model imperfections, making true performance worse while appearing to optimize the proxy.



Problem 3

Here are the quantitative results:

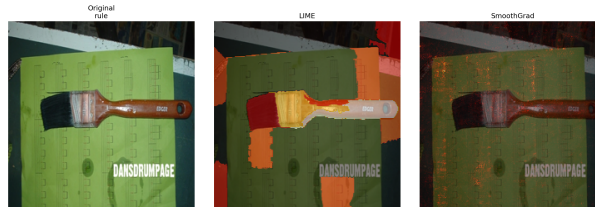
Image	Prediction	Kendall's Tau	Spearman's Rho
piggy_bank	pillow	0.1000	0.1325
paintbrush	rule	0.0060	0.0083
mousetrap	drumstick	0.1961	0.2566
peacock	peacock	0.1157	0.1592
mountain_bike	mountain_bike	0.0047	0.0070

Piggy Bank (misclassified as pillow):



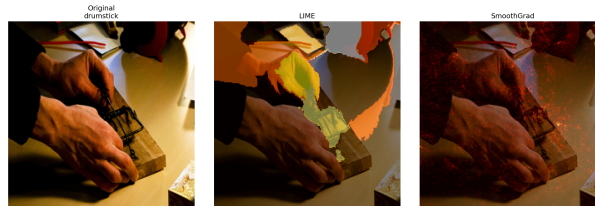
The misclassification likely occurs because the rounded shape and smooth texture resemble a pillow. Both methods highlight the central body rather than distinctive piggy bank features.

Paintbrush (misclassified as rule):



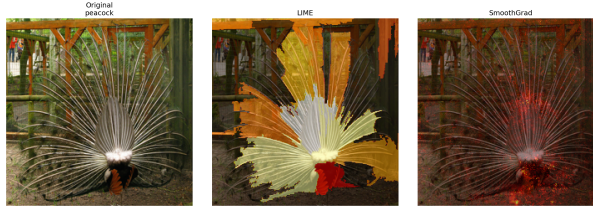
LIME focuses on the handle region. SmoothGrad highlights brush bristles and edges. The elongated straight shape may cause confusion with a ruler.

Mousetrap (misclassified as drumstick):



LIME highlights the wooden base and human hand. SmoothGrad shows edges and metal components. The cylindrical wooden piece may resemble a drumstick, and humans commonly hold drumsticks.

Peacock (correct):



LIME identifies large feather regions. SmoothGrad highlights detailed feather patterns and the bird's body. Both correctly focus on distinctive peacock features.

Mountain Bike (correct):



LIME highlights wheel and frame regions. SmoothGrad emphasizes edges of wheels, frame, and handlebars. Both correctly identify key bicycle components.

As we can see from the examples, LIME produces blocky, region-based explanations that are easily interpretable. SmoothGrad provides fine-grained, pixel-level importance with smoother gradients but can be noisier. LIME is better for understanding which image regions matter, while SmoothGrad better shows detailed edges and textures, though the noiseness of SmoothGrad makes it more difficult. Low positive correlations (0.00-0.20) indicate methods highlight different aspects. LIME focuses on coherent regions while SmoothGrad emphasizes gradients/edges. This signals that these two methods should be complementary, not single bellwethers.